1. SARTA Analytical Jacobians (Weighting Functions)

In satellite data assimilation and atmospheric profiles, knowing the final brightness temperature is only half the battle. Scientists also need to know **how much** that temperature will change if a specific layer of the atmosphere warms up or dries out. This sensitivity matrix is called a **Jacobian** (K).

Because SARTA uses smooth, differentiable linear regression polynomials to calculate layer optical depths (τ), it can calculate these sensitivities analytically using the **Calculus Chain Rule**. This avoids the slow process of manually perturbing every single level one by one.

Temperature Jacobians (K_T)

The temperature Jacobian shows how sensitive a channel's top-of-atmosphere radiance (I_{TOA}) is to a temperature change (∂T) at layer L:

$$K_T(L) = \frac{\partial I_{TOA}}{\partial T(L)}$$

SARTA splits this derivative into two main physical components:

1. **The Shift in Planck Emission**: A change in the layer's internal thermal radiation source function ($\partial B_L / \partial T_L$).
2. **The Shift in Gas Transmittance**: A change in the layer's optical thickness ($\partial \tau_L / \partial T_L$), since molecular absorption cross-sections scale with temperature. SARTA finds this instantly by differentiating the fixed and variable gas polynomial lines directly against the temperature term (ΔT).

Water Vapor and Trace Gas Jacobians (K_q)

Gas Jacobians track sensitivity to changes in composition (like water vapor content, q):

$$K_q(L) = \frac{\partial I_{TOA}}{\partial q(L)}$$

Because changing a layer's gas concentration alters both the energy emitted *by* that layer and how much light from below can pass *through* it, SARTA calculates this by differentiating its gas regression strings against the scaling ratio (q/q_{ref}). This gives researchers a direct map of exactly where a satellite channel is looking in the atmosphere.

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2. Surface Reflection and Downwelling Radiation

The lower boundary of SARTA's 100-layer grid does not just emit energy; it also acts as a mirror that reflects thermal energy back up to space. To accurately model this, SARTA accounts for **surface emissivity (ε)** and its complement, **surface reflectivity ($1 - \varepsilon$)**.

```
                      [ To Space ]
                            ▲
                            |  (Upwelling Stream)
   ┌────────────────────────┴────────────────────────────┐
   |          Atmospheric Path                           |
   └────────────────────────▲────────────────────────────┘
                            |
          ┌─────────────────┴─────────────────┐
          |  Total Radiance at Surface        |
          └─────────────────▲─────────────────┘
                            |
          ┌─────────────────┴─────────────────┐
          |    [ Earth Surface Boundary ]     |
          └───────▲───────────────────▲───────┘
                  |                   |
    (Surface Emission)     (Reflected Downwelling)
     ε * B(λ, T_skin)        (1 - ε) * F_down
```

To calculate the total upwelling boundary radiance at the surface, SARTA combines two main components:

A. Thermal Surface Emission

The background energy naturally radiated by the planet's surface:

$$I_{emit} = \varepsilon(\lambda) \cdot B(\lambda, T_{skin})$$

Over oceans, SARTA calculates the spectral emissivity (ε) dynamically based on wind speed to account for wave roughness. Over land, it uses global emissivity maps derived from satellites like MODIS.

B. Reflected Downwelling Atmospheric Radiance

The atmosphere itself radiates thermal energy downward toward the ground. The surface reflects a fraction of this energy back up into the satellite's line of sight:

$$I_{reflect} = (1 - \varepsilon(\lambda)) \cdot F_{down}(\lambda)$$

Where F_{down} is the total downwelling atmospheric thermal flux hitting the ground. SARTA approximates this downward stream quickly during its vertical pass to ensure it does not slow down operational execution speeds.

C. Reflected Solar Specular Flare (Shortwave Infrared Channels Only)

For shortwave channels (wavelengths shorter than 4 μm), solar radiation becomes a significant factor during daytime passes. SARTA models the solar beam traveling down through the atmosphere, bouncing off the surface, and traveling back up to the satellite. It adjusts this calculation based on the **Solar Zenith Angle** and the surface bidirectional reflectance distribution function (BRDF).

Would you like to see how SARTA applies these techniques to specific satellite instruments like **AIRS** or **CrIS**, or look at how it compares to slower **line-by-line forward models (LBLRTM)**?